

Maryam Jabbari

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Professional Summary

M.Sc. student in Bioinformatics with a B.Sc. in Computer Science and research experience in retrieval-augmented learning, knowledge distillation, and efficient 3D medical image analysis. My current work spans retrieval-augmented protein language models and efficient deep learning for volumetric medical image segmentation. I am particularly interested in multimodal representation learning, foundation models, and efficient deep learning for biomedical data.

Education

2025–Present	M.Sc. in Bioinformatics <i>Amirkabir University of Technology (Tehran Polytechnic), Iran</i> • Thesis: Retrieval-Augmented Protein Language Models for Multi-Label Antimicrobial Peptide Classification • Selected Courses: Analysis of Protein and Nucleic Acid Sequences (20/20) • Machine Learning (19.75/20) • Modeling and Prediction of Macromolecular Structures (19.5/20) • Special Topics: Deep Learning and Computational Physics (19.25/20) GPA: 19.26/20
2021–2025	B.Sc. in Computer Science <i>Amirkabir University of Technology (Tehran Polytechnic), Iran</i> • Thesis: Feature-Based Knowledge Distillation with CBAM Attention Grade: 20/20 Supervisor: Dr. F. Shakeri • Selected Courses: Data Mining (20/20) • Design and Analysis of Algorithms (20/20) • Linear Optimization (20/20) • Fundamental of Bioinformatics (18.92/20) • Neural Networks (18.83/20) • Artificial Intelligence (18.75/20) GPA: 18.87/20

Research Interests

- Multimodal & Representation Learning
- Foundation Models & Retrieval-Augmented Learning
- Efficient Deep Learning & Model Adaptation
- 3D Medical Image Analysis

Research Experience

Jun 2026–Present	Retrieval-Augmented Protein Language Models for Multi-Label Antimicrobial Peptide Classification <i>M.Sc. Thesis under Dr. F. Zare and Dr. Z. Ghorbanali, Amirkabir University of Technology</i> • Adapting a retrieval-augmented protein language model to peptide sequences for multi-label antimicrobial function classification; comparing retrieval strategies with non-retrieval and frozen-retrieval baselines. • Designing controlled experiments and systematic evaluation protocols to study how retrieved sequence information affects learned representations and downstream classification.
Sep 2024–Present	Efficient 3D Medical Image Segmentation through Joint Architectural and Pipeline Design <i>Research Project under Dr. F. Shakeri Iran National Science Foundation (INSF) Manuscript in Preparation</i> • Contributing to the design and evaluation of a MobileNet-based 3D CNN with inverted bottleneck blocks for volumetric medical image segmentation. • Investigating input-adaptive dynamic convolutions, target-spacing, and stem-stride design for efficient volumetric inference and contributing to experimental analysis and manuscript preparation.
Jun 2024–May 2025	Feature-Based Knowledge Distillation with CBAM Attention <i>B.Sc. Thesis under Dr. F. Shakeri, Amirkabir University of Technology</i> • Developed a teacher–student knowledge-distillation framework on CIFAR-100 using a ResNet-56 teacher and ResNet-20 student, with CBAM-guided intermediate feature matching. • Combined feature-level transfer with conventional logit distillation and evaluated the effect of attention-guided knowledge transfer on student-model performance. Code
Oct 2022–Dec 2023	Autonomous Vehicle Perception in Simulation <i>Robotics Team Member, Robotics Lab, Amirkabir University of Technology</i> • Developed and tested computer-vision components in the AVIS Engine simulator using Python, OpenCV, NumPy, and YOLO-based obstacle detection. • Contributed to the team awarded 1st Place in the Simulation League at the IRAN FIRA RoboWorld Cup Open.

Selected Projects

Jul 2026–Aug 2026

Multilabel Antimicrobial Peptide Classification Benchmark

- Built a controlled five-label AMP benchmark on ESCAPE using the reciprocal Fold1/Fold2 training protocol and an independent test set.
- Compared seven frozen protein/peptide language-model encoders with six biological sequence representations under a shared flat MLP; also evaluated the biological representations with Random Forest classifiers.
- Implemented overlap auditing, leakage-controlled splits, embedding caching, reciprocal-fold ensembling, and evaluation under severe multilabel imbalance using mAP, macro/micro F1, balanced accuracy, and per-label diagnostics.

Stack: PyTorch · Transformers · scikit-learn · NumPy

Feb 2025–Jun 2025

[Code](#)

Image Reconstruction via Regularized Inverse Problems

- Modeled image degradation as the linear inverse problem $b = Ax + \eta$ using Gaussian blur, motion blur, and additive noise, with the spatial forward operator represented as a SciPy LinearOperator.
- Compared unregularized LSQR, zero-order Tikhonov regularization, Wiener filtering, and Richardson–Lucy deconvolution.
- Analyzed reconstruction quality and parameter sensitivity using PSNR and SSIM with recorded parameter sweeps.

Stack: NumPy · SciPy · scikit-image · Numerical Optimization

[Code](#)

Dimensionality Reduction for scRNA-seq Data

- Built dataset-specific preprocessing pipelines for Deng (GSE45719), Chu (GSE75748), and a filtered PBMC68k subset, including filtering, normalization, log transformation, HVG selection, and feature scaling.
- Compared PCA, t-SNE, UMAP, and a project-specific PyTorch VAE using low-dimensional visualizations and higher-dimensional representations for downstream clustering.
- Evaluated K-means clustering using ARI, AMI, and Silhouette Score and documented methodological differences from the reference benchmark.

Stack: PyTorch · scikit-learn · UMAP · Scanpy/scVelo

Feb 2024–Jun 2024

[Code](#)

Kolmogorov–Arnold Network

- Implemented a KAN from scratch in PyTorch where each connection learns a univariate function using cubic B-spline bases and a learnable SiLU residual component.
- Implemented cubic B-spline evaluation with iterative Cox–de Boor dynamic programming, trainable basis coefficients, and knot grids registered as PyTorch buffers.
- Benchmarked against width- and parameter-matched MLP baselines and added correctness, gradient-flow, and learned edge-function visualization tests.

Stack: PyTorch · NumPy · scikit-learn · B-Splines

[Code](#)

Autograd Engine and Neural Network from Scratch

- Implemented a scalar reverse-mode automatic differentiation engine with computational-graph construction, differentiable scalar operations, topological backpropagation, and gradient accumulation.
- Built Neuron, Layer, and MLP abstractions on the custom engine, including nonlinear activations, MSE loss, gradient-descent updates, and graph visualization.

Stack: Python · Reverse-Mode Autodiff · Backpropagation · Graphviz

Honors & Awards

2026	Ranked 1st in the M.Sc. entering cohort based on cumulative GPA, Amirkabir University of Technology.
2024	Direct admission to the M.Sc. program through the Exceptional Talents pathway , without the national graduate entrance examination and with tuition-free admission. Ranked in the top 10% of the B.Sc. entering cohort based on final cumulative GPA, Amirkabir University of Technology.
2023	1st Place, Simulation League – IRAN FIRA RoboWorld Cup Open , Robotics Team MCAC, Amirkabir University of Technology.

Teaching Assistant Experience

Sep 2026–Present	Bioinformatics (M.Sc.) <u>Dr. F. Zare</u> Preparing and refining lecture materials and contributing to graduate-level assignments and the course project.
Sep 2025–Jan 2026	Deep Learning (B.Sc.) <u>Dr. F. Shakeri</u> - Contributed to preparing and refining course materials and evaluating project presentations, reports, code, and final deliverables. Numerical Linear Algebra (B.Sc.) <u>Dr. M. Dehghan</u> - Contributed to homework/final-project design and evaluation; led the exam-question design subgroup and reviewed questions and solution keys.
Feb 2025–Jan 2026	Artificial Intelligence and Workshop (B.Sc.) <u>Prof. M. Ghatee</u> Contributed to assignment and course-project design and evaluated student presentations and project deliverables.
Feb 2025–Jun 2025	Foundations of Numerical Analysis (B.Sc.) <u>Dr. F. Shakeri</u> - Contributed to problem-set design and evaluated student submissions. Operating Systems (B.Sc.) <u>Dr. Z. Ghorbanali</u> - Prepared course materials, contributed to assignment/project design, and graded student submissions with feedback.
Feb 2024–Jun 2024	Advanced Programming and Workshop (B.Sc.) <u>Prof. M. Ghatee</u> - Contributed to programming exercises/projects and led interactive problem-solving and workshop sessions. Design and Analysis of Algorithms (B.Sc.) <u>Dr. A. Dolati</u> - Contributed to assignment-question design and evaluated student submissions.
Sep 2023–Jan 2024	Foundations of Numerical Analysis (B.Sc.) <u>Dr. M. Abbaszadeh</u> - Led problem-solving and tutorial sessions to review course material and address student questions. Introduction to the Theory of Computation (B.Sc.) <u>Dr. N. Roshandel Tavana</u> - Led problem-solving and tutorial sessions and evaluated student assignments.

Skills

Programming	Python, C++, Java, C, Rust
ML & Deep Learning	PyTorch, Hugging Face Transformers, scikit-learn
Scientific & Image Computing	NumPy, SciPy, OpenCV, scikit-image, pandas
Bioinformatics	Scanpy, scVelo, AlphaFold2, PyMOL
Tools	Git, GitHub, L ^A T _E X
Languages	English (CEFR C1); Persian (Native)

Volunteering

Jul 2026 – Present	Student Scientific Magazine “Halghe” Amirkabir University of Technology Contributed to scientific editing and editorial review, and suggested relevant scientific articles and books for publication and discussion.
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